

Likelihood.

$$L(\pi) = \pi^k (1 - \pi)^{N-k}.$$

1. **Take logs.** (Use $\log(ab) = \log a + \log b$ and $\log(x^c) = c \log x$.)

$$\begin{aligned}\log L(\pi) &= \log(\pi^k (1 - \pi)^{N-k}) \\ &= \log(\pi^k) + \log((1 - \pi)^{N-k}) \\ &= k \log \pi + (N - k) \log(1 - \pi).\end{aligned}$$

2. **Differentiate $\log L(\pi)$ with respect to π .**

$$\begin{aligned}\frac{d \log L(\pi)}{d\pi} &= k \frac{d \log \pi}{d\pi} + (N - k) \frac{d \log(1 - \pi)}{d\pi} \\ &= k \left(\frac{1}{\pi} \right) + (N - k) \left(\frac{1}{1 - \pi} \frac{d(1 - \pi)}{d\pi} \right) \\ &= \frac{k}{\pi} - \frac{N - k}{1 - \pi}.\end{aligned}$$

3. **Set the derivative to zero at $\pi = \hat{\pi}$ and solve.**

$$\begin{aligned}\left. \frac{d \log L(\pi)}{d\pi} \right|_{\pi = \hat{\pi}} &= \frac{k}{\hat{\pi}} - \frac{N - k}{1 - \hat{\pi}} = 0. \\ \frac{k}{\hat{\pi}} &= \frac{N - k}{1 - \hat{\pi}} \iff k(1 - \hat{\pi}) = (N - k)\hat{\pi} \\ \iff k &= N\hat{\pi} \implies \hat{\pi} = \frac{k}{N} = \frac{1}{N} \sum_{i=1}^N y_i.\end{aligned}$$

4. **Second-order condition (concavity).** Need $\frac{d^2 \log L(\pi)}{d\pi^2} < 0$.

$$\frac{d^2 \log L(\pi)}{d\pi^2} = -\frac{k}{\pi^2} - \frac{N - k}{(1 - \pi)^2} < 0 \quad (0 < \pi < 1).$$

Thus $\log L(\pi)$ is strictly concave, and the critical point $\hat{\pi} = k/N$ is the unique global maximizer.